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SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

This month in Earth we explore fibre seismology, we learn about the origin of thermodynamics, in Space Age we delve into dark energy data dump, and more.

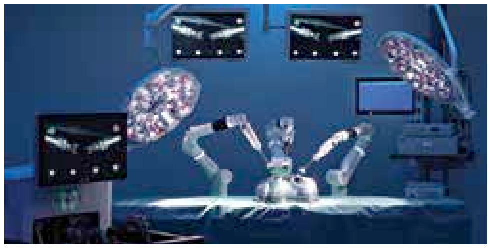
M87 imaged, again

The same team of researchers from the Event Horizon Telescope that released the first image of a black hole, have now released an image of magnetic fields along the edge of M87, that shows the polarisation and direction of the light around the black hole shadow. **http://dgit.in/m87shadow**

WHAT'S NEW

What will 6G look like?

5G is yet to be rolled out around the world but the researchers at the Technical University of Munich are already working on the criteria for the 6G standard, and are aiming to achieve 99.999999999% reliability, and as little latency as possible, not focusing on the speed so much, which is still expected to be around a terabyte per second. This is important for critical applications such as telesurgery, where lives will be on the line. Prioritising reliability and low latency will be required for other critical applications that are expected to become commonplace, including self driving vehicles, as well as robots in childcare and healthcare. Another criteria for 6G is that the



researchers want to make the network itself intelligent, allowing it to react dynamically and in realtime to optimise itself. An idea researchers are exploring is to make a digital twin of the entire network and all its components, allowing them to run simulations to see how the network will behave under pressure. **http://dgit.in/tum6g**

Underwater drones to pick up litter

The SeaClear project has developed the first underwater drones for picking up litter from the oceans. Currently, most of the cleaning from the seafloor bottom is done manually by divers. The SeaClear project uses a team of autonomous underwater drones, that are connected to a vessel that acts as a command and control hub for the entire fleet, powering the power and computation as well. The autonomous underwater

vehicles then search, identify and collect marine litter. Initial scans of an area are conducted by both the hub vessel as well as scouting drones. Once the litter in the area is mapped, a collector bot uses custom made graspers to grab the litter. The entire system relies heavily on AI, as the robots have to be smart enough to differentiate between litter and life forms. While the scout and grabber drones are connected by underwater cables to the hub ship, the drone can be freeflying, or tethered. The kicker here is that the hub vehicle is an autonomous surface vehicle, so this system consists of ASVs, UAVs and AUVs. **http://dgit.in/seaclear**

Underground farming

At an unused air raid shelter in England, researchers have set up a smart and energy-efficient underground farm that has the potential to revolutionise urban farming. These subterranean farms in the future can potentially be used to feed the cities, from under the cities. **http://dgit.in/unfa**

No Puffin food

Researchers using tiny GPS trackers to observe the feeding patterns and movements of Puffins in the northeast atlantic found that the populations of these birds were getting decimated because of a lack of sufficient prey, and that puffin chicks were just starving to death. **http://dgit.in/puffinchicks**

Quantum brilliance

Researchers from the Australian National University are in the final stages of installing a small, lunchbox sized device which happens to be the world's first quantum computer that works at room temperatures, which also uses diamonds in a unique way to operate. **http://dgit.in/quantumbrilliance**